Ibrahim Khaliq

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EDUCATION

University of Richmond | Richmond, VA May 2028
B.S., Computer Science & Mathematics | Minor in Data Science and Statistics
Coursework: Data Structures, Software Systems, Machine Learning, Linear Algebra, Computer Organization
Leadership: ACM Events Chair; Pakistani Student Assoc. Treasurer; Muslim Life Coordinator; VITA Tax Volunteer

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C++, SQL, JavaScript, HTML/CSS
Frameworks/Libraries: FastAPI, Pydantic, Pandas, NumPy, scikit-learn, OpenCV, Streamlit
Tools/Platforms: Git/GitHub, Docker, AWS, PostgreSQL, SQL Server, Linux, Android Studio, Jupyter
Concepts: REST APIs, ETL, ML pipelines, vector search, testing (pytest, JUnit), CI/CD

EXPERIENCE

Undergraduate Research Assistant | SASSEMA, University of Richmond Jan 2026 – Present
- Led Java-based Android port of ADHD research app with onboarding, recovery logic, and 20+ questionnaires (200+ fields).
- Built AlarmManager reminders and accelerometer/gyroscope sensing (25–50 Hz) for field data capture.
- Integrated five Health Connect data types via Kotlin/Java; fixed SDK check blocking integration on Android 16.
- Built WorkManager/JSONL pipeline with Wi-Fi-gated uploads and retries; 45 QA cases caught 7 bugs.

CLAII Fellow | Center for Liberal Arts and AI, University of Richmond Aug 2026 – Present
- Building LLM pipeline turning CHR (digital humanities journal) articles into posts for Bluesky, X, LinkedIn, Instagram.
- Designed six-stage system (ingestion, LLM drafting, editor approval queue, delivery) with mandatory human review.
- Built few-shot prompts from 24 editor-written posts (eval baseline); automated Bluesky pipeline after comparing 4 APIs.

Software Engineering Intern | Express Tax and Accounting Aug 2025 – Sep 2025
- Tuned SQL queries with composite indexes across PostgreSQL/SQL Server, cutting report generation time by 40%.
- Added Pydantic validation and error handling, eliminating 200+ monthly duplicate-entry errors.
- Reconciled 5,000+ monthly ledger transactions from three sources using Python/Pandas; cut manual entry by 35%.
- Built tax-document ingestion and normalization ETL, saving auditors 10+ hours weekly.

Innovation Fellow | Richmond Innovation Fellows Jan 2025 – Present
- Selected 1 of 8 from 100 applicants; conducted 15+ stakeholder interviews and evaluated 20+ platforms.
- Prototyped dining-hall fixes with a 3-person team; pitched one improvement to the Head of Dining and faculty.

Resident Assistant | Residence Life, University of Richmond Aug 2025 – Present
- Support 30+ residents through conflict resolution, referrals and community programs with an eight-person RA team.

PROJECTS

AI GitHub Repository Assistant | Python, FastAPI, Ollama, Docker
- Built LLM-powered FastAPI service to clone/index repos, summarize code and answer architecture questions.
- Added hybrid keyword + embedding search, raising top-5 retrieval accuracy from 60% to 85% on a 20-question eval.
- Parallelized indexing and added commit-aware caching, cutting re-index time on a large repo from 5 min to 1 min.

Real-Time Facial Recognition | Python, OpenCV, dlib
- Developed webcam recognition at 15+ FPS that enrolls new identities without model retraining.
- Matched live face embeddings against registered identities and automated logging timestamped per-session attendance to CSV.

ADDITIONAL INFORMATION

Spoken Languages: English, Urdu, Punjabi (fluent in speaking, reading and writing).
Interests: Cricket, Chess, Soccer, Reading, Anime